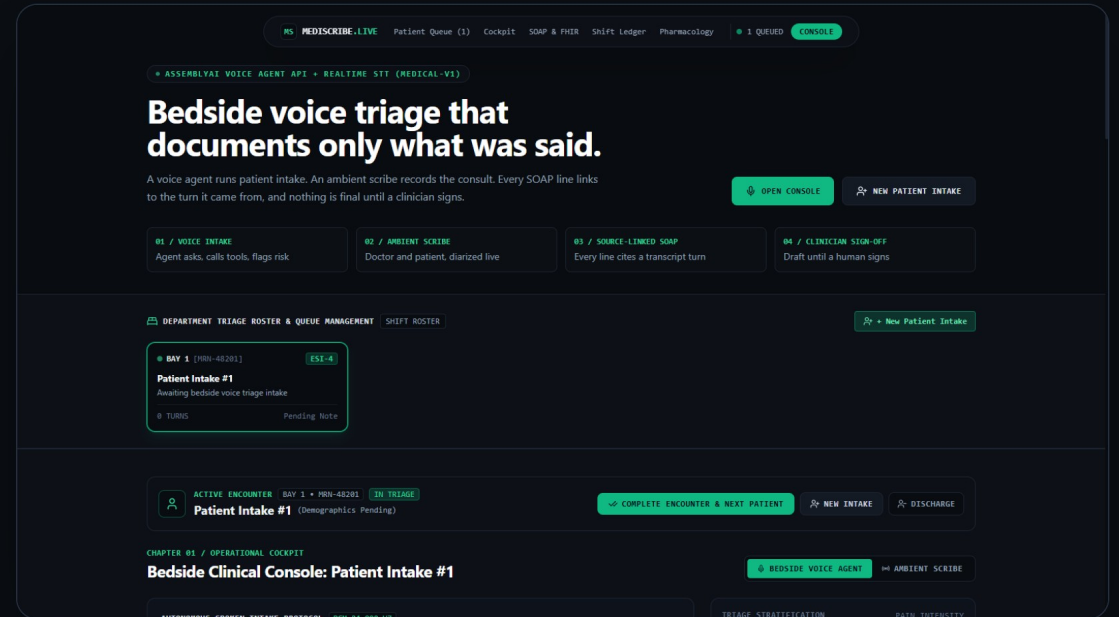


ASSEMBLYAI VOICE AGENT HACKATHON

MediScribe Live

Bedside voice triage that documents only what was said.



mediscribe.rhasan.online · github.com/Daddy-Ousen/mediscribe-live

THE PROBLEM

Notes take the shift. Unchecked AI notes take the trust.

49%

of physicians' time in the clinic went to EHR and desk work. Only 27% went to direct time with patients.

Sinsky et al., Annals of Internal Medicine, 2016

AI scribes write fast. Clinicians still cannot trust them.

A generated note can add an allergy, a dose or a diagnosis that nobody said.

If a line has no source, the clinician must re-check the whole note. The time saving is lost.

THE WORKFLOW

Four steps, one patient chart

01

Voice intake

A voice agent talks with the patient and calls clinical tools.

Voice Agent API

02

Ambient scribe

Records the consult.
Separates doctor and patient.

Streaming STT

03

Source-linked SOAP

Every line cites the transcript turn it came from.

LLM Gateway

04

Clinician sign-off

Draft until a named clinician signs. Exports HL7 FHIR R4.

FHIR Bundle

Both voice modes write to the same chart: name, complaint, vitals, medicines, allergies and ESI acuity.

The voice agent runs bedside intake

Full-duplex talk over one WebSocket. The patient can interrupt, and the agent stops speaking at once.

While it talks, the agent calls tools. The chart and the alerts update live.

Latency is measured, not claimed

The panel shows the time from end of speech to the first agent audio: last value and median.

`record_patient_intake`

Name, age, complaint, pain score and vitals go into the chart.

`check_drug_interaction`

Example: warfarin + aspirin raises a bleeding-risk alert.

`flag_critical_vital`

An unsafe vital sign flags the patient and raises the ESI level.

The scribe hears who said what

- Universal-3.5 Pro with the medical-v1 domain for drug and body terms
- Speaker labels split Doctor and Patient; one click swaps them
- Final turns fill the chart: name, complaint, vitals, medicines

REALTIME AMBIENT CONSULTATION SCRIBE

PCM 16,000 HZ • medical-v1

START AMBIENT SCRIBE

Continuous doctor-patient transcription with medical domain acoustic tuning for Maria Lopez.

AUDIO DSP TELEMETRY

16,000 HZ PCM16 MONO DISCONNECTED

-48 dB

-36 dB

-24 dB

-12 dB

-6 dB

0 dB CLIP

CARRIER: OFFLINE

MIC: IDLE

BARGE-IN: ARMED

IDENTIFYING SPEAKER... morning. 5:33:07 AM

DOCTOR What brings you in today? 5:33:09 AM

PATIENT My name is Maria Lopez. 5:33:11 AM

PATIENT I have had a bad headache for 2 days, and light hurts my eyes. 5:33:16 AM

Maria Lopez • 10 consultation turns recorded

SNAP DOCTOR / PATIENT

COMPILE SOAP DOCUMENTATION

CLINICAL EVENT AUDIT LEDGER • MARIA LOPEZ

0 DISPATCHED

Audited tool invocations for Maria Lopez (check_drug_interaction, flag_critical_vital) will render here in real time.

TRIAGE STRATIFICATION

PAIN INTENSITY

ESI-5 NON-URGENT -- / 10

CHIEF CLINICAL COMPLAINT

I have had a bad headache for 2 days, and light hurts my eyes.

PHYSIOLOGICAL TELEMETRY

LIVE TELEMETRY

BLOOD PRESSURE 150/95 mmHg

HEART RATE -- bpm

SPO2 --%

RESPIRATORY RATE --/min

TEMPERATURE --°F

ACTIVE CHART DOSSIER

Maria Lopez

RECONCILED HOME MEDICATIONS

tramadol RX-ACTIVE

sertraline RX-ACTIVE

RECORD IDENTIFIER: MRN-48201

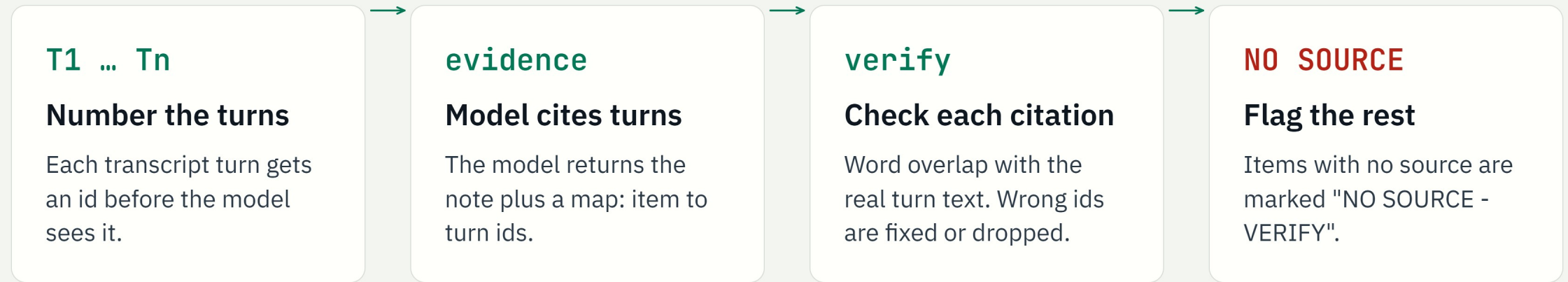
UNIT / LOCATION: BAY 1 • TRAUMA TELEMETRY

ENCOUNTER TIME: 12:00:00

Every SOAP line points to the words that were said.

HOW CITATIONS ARE CHECKED

The server checks the model's sources



In the note, each T-chip opens the exact quote. The clinician checks a line in one click.

Maria Lopez

SOAP-650088

SIGNED

2026-09-21 • Signed by Dr. Demo Clinician, MD at 9/22/2026, 5:35:08 AM • qwen3.5-4b-32k-fast

</> HL7 FHIR R4

Copy Note

🔒

SOURCE-LINKED: 9/9 items

Click a T-number to see what was said.

🔗 SOURCE TURNS

T7 Patient "I get hives."

I. SUBJECTIVE

Chief Complaint: Headache for 2 days with light sensitivity T4

History of Present Illness: Patient reported a bad headache for 2 days and noted that light hurts her eyes. She took tramadol last night for the pain. T4

ALLERGIES

Sulfa drugs (hives) T7

CURRENT MEDICATIONS

Sertraline, taken every day; Tramadol, taken last night T9

II. OBJECTIVE

Blood Pressure

150/95 mmHg

Heart Rate

Not recorded during encounter

SpO2

Not recorded during encounter

Respiratory Rate

Not recorded during encounter

Temperature

Not recorded during encounter

PHYSICAL EXAMINATION

• Neck is supple T10

DIAGNOSTIC RESULTS

• Blood pressure 150/95 T10

III. ASSESSMENT

Migraine T10

Patient reported 2 day headache with photophobia (T4-T5). Clinician noted physical exam and vitals (T10) and stated diagnosis of migraine (T10). Triage tier: ESI-5 Non-Urgent. Medication interaction alert: tramadol + sertraline (Major Warning).

AI-SUGGESTED DIFFERENTIAL (UNCONFIRMED): Migraine [643.9]

IV. PLAN

Ketorolac (dose (specific amount not stated)) T10

give to patient

DIAGNOSTIC ORDERS

None stated in dialogue

REAL TEST RUN

9 of 9 items linked to a source

Click T7 and the note shows the quote: "I get hives."

Vitals nobody said stay "Not recorded during encounter".

After sign-off the note shows SIGNED, the clinician's name and the time.

GUARDRAILS

Rules the note cannot break

If...	The note does this
A topic was not discussed	Says "Not discussed during encounter"
No allergy was stated	Leaves it open. Never defaults to NKDA
No diagnosis was stated	Shows no diagnosis or ICD code
The AI suggests a differential	Labels it "AI-suggested (unconfirmed)"
The LLM call fails	Makes no note. No template fallback
Nobody has signed yet	Stays DRAFT; FHIR Composition is "preliminary"
A clinician signs	FHIR Composition becomes "final" with an attester

ARCHITECTURE

The API key never reaches the browser

Browser

Next.js 14 + React. Web Audio:
mic to PCM16, fixed 100
ms frames.

direct WebSockets
→
short-lived token

Voice Agent API

24 kHz PCM · tool calls · barge-in

Streaming STT v3

16 kHz · universal-3-5-pro · medical-v1

LLM Gateway, via our server

SOAP + evidence map · citation check

Next.js API routes mint the tokens and call the LLM. Deployed on Vercel.

Faster notes that can pass an audit

Who it is for

- Emergency departments and urgent care
- Clinicians: less typing, one-click checks
- Hospitals: every line has an audit trail
- EHR teams: standard HL7 FHIR R4 output

Next steps

- A stronger LLM for SOAP synthesis
- A full drug database in place of the demo list
- HIPAA-ready hosting and a pilot with one clinic
- Write signed notes back into the EHR



Try MediScribe Live

mediscribe.rhasan.online

github.com/Daddy-0usen/mediscribe-live

Built with AssemblyAI Voice Agent API, Streaming STT and LLM Gateway. MIT License.